Individual Tax-Saving Advisor (RAG w/ Milvus Lite and Base Model w/ GPT-oss-20b)

This notebook:

1. Creates the Milvus Lite and Hybrid RRF-ready collections, instructions, and few-shots
2. Checks and creates/ingests into the IRS-ready Hybrid dense+BM25 collection (or reindexes) per `MILVUS_FORCE_REINDEX`
3. Loads the base model (`openai/gpt-oss-20b`) and tokenizer
4. Defines retrieval helpers (dense, BM25, hybrid RRF) and a few-shot builder
5. Defines a strict JSON-only system prompt for tax-saving advice, requiring `tax_year`, `retrieval_queries`, `filing_status`, `assumptions`, `steps`, `risks`, `citations`, `missing_info`, and `notes_for_user`
6. Supports multi-turn **conversation** with state tracking for user-provided facts
7. Runs a **single-turn** example and a **multi-turn** conversation demo
8. Includes optional cleanup to free GPU memory when finished

Important notes:

- Tailored for tax-saving strategies
- Not tax filing or legal advice

0) Install dependencies

```
!pip -q install -U "pymilvus[milvus-lite]" sentence-transformers "accelerate>=0.26.0"
!pip -q install -U bitsandbytes>=0.43.1
!pip -q install -U "transformers>=4.51.3,<4.57"
"tokenizers>=0.22,<0.23"
"huggingface-hub>=0.34,<1.0"
!mkdir -p /content/milvus
!ls -sh /usr/bin/{curl,tar} | head -n 2
```

```
bash: cannot set terminal process group (-1): Inappropriate ioctl for device
bash: no job control in this shell
Requirement already satisfied: transformers!=4.51.0,< in /usr/local/lib/python3.12/dist-packages (4.51.3)
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Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (2024.11.6)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (2.32.3)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (4.67.1)
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Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /usr/local/lib/python3.12/dist-packages (0.34.4)
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Requirement already satisfied: urllib3<1.26,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (1.25.11)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (2025.8.3)
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Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1)
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Requirement already satisfied: 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    - ssh_port=22,ssh_flag=true
  - ssh_port=22,ssh_flag=true
  - ssh_port=22,ssh_flag=true
  
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```
    print("Wrong:", len(sub))
    for _, r in sub.iterrows():
        print("\nQ:", r["question"])
        print("Gold:", r["gold"], "Pred:", r["pred"])
```

```
============================================================================
Topic: gift_estate_and_charitable
Wrong: 2

Q: John sells his house to his adult child for $375,000 cash. At the time of the sale, the house has afair market value of $450,000, a mortgageof $200,000, and an ad
justed basis of $250,000. John pays off the mortgage. Before the gift tax annual exclusion is taken into account,what amount is deemed to bea gift by John?
Gold: A Pred: D

Q: Which of the following transactions is subjectto the gift tax beforethe gift tax annual exclusion is taken into account?
Gold: A Pred: C

============================================================================
Topic: passive_activity_and_at_risk
Wrong: 4

Q: In the current tax year, Charlie reported the following: Wages $ 50,000 Taxable interest 1,500 Dividends 3,000 Partnership X loss (15,000) Partnership Z income 3,
000 Charlie does not materially participate in either partnership X or Z. In partnership X,he hasa tax basis of $10,000and an at-risk basis of $7,000.In partnership
Z his tax basis is $15,000, and his at-risk basis is $14,000. What isCharlie's suspendedloss carryforward resultingfrom the tax basis, at-risk basis, and passiveloss
limitations for partnership X?
Gold: B Pred: D

Q: Thomas, asingletaxpayer, is a partner in partnership A and partnership B.Thomas materially participates in partnershipA but does notmaterially participate in part
nership B. This year, Thomas's shareofthepartnershiplossinpartnershipAis$12,000andhisshare of the partnership incomein partnership B is$4,000. Other important inform
ation: Tax basis in partnership A $10,000 At-risk basis in partnership A 9,000 Tax basis in partnership B 20,000 At-risk basis in partnership B 19,000 What isThomas'
s suspended loss carryforward for partnership A due to the at-risk basis?
Gold: C Pred: B

Q: An individual taxpayer has ownershipinterests inthreeseparate passive activities. In Year 2, the passive activities generated the followingincome or losses: Activ
ity 1 ($40,000) loss; Activity 2 $90,000income;Activity3($80,000) loss; net passive loss ($30,000). What amount of the net passive loss, ifany, is allocated to Activ
ity 3 as a suspended passive activity loss?
Gold: C Pred: B

Q: An individual reportednet incomeof $110,000 from business activitiesand anet loss from a passive rental activity of $40,000 for the current year. What amountof pa
ssive loss, ifany, may be deducted for the current year?
Gold: D Pred: C

============================================================================
Topic: property_transactions
Wrong: 1

Q: Chad owned an officebuilding that was destroyed in a tornado.The adjusted basis of the building at the time was $890,000. After the deductible, Chad received an i
nsurance check for $950,000. He used $900,000 of the insuranceproceeds to purchasea new building that sameyear. How much is Chad's recognized gain, and what is his b
asis inthe new building? Recognized Gain New Basis
Gold: C Pred: D

============================================================================
Topic: retirement_and_financial_planning
Wrong: 1

Q: During a tax year, a 45-year-old taxpayer earned $50,000 at a pharmacy that provided access to a Roth 401(k) plan. The taxpayer elected to contributethe maximum a
nnual amount allowed into the Roth 401(k) plan. Thetaxpayer also earned $60,000duringthe year at ahospital pharmacy. The taxpayer has worked for ten years at the hos
pital pharmacy, whichgives employees access to a 403(b) plan. Which of the following statements is correct regarding the limitations on employee contributions to the
taxpayer's 403(b) account for this tax year?
Gold: A Pred: B
```

In [80]:

Comparing with base LLM

In [81]:

```
import torch
from tqdm.auto import tqdm
import pandas as pd

def answer_mcq_raw_model(question: str, options: dict) -> str | None:
    """Raw model MCQ -- no retrieval, computation-aware prompt."""
    options_text = "\n".join([f"{k}. {v}" for k, v in options.items()])

    prompt = f"""Answer this multiple-choice U.S. tax question.

Question:
{question}

Options:
{options_text}

Instructions:
1. Identify what tax rule or computation the question is testing.
2. If the question involves numbers, show each calculation step explicitly.
3. After completing your analysis, compare your result to each option A, B, C, D.
4. Pick the option that matches your computation.

ANSWER: """

    raw = _generate_mcq(prompt, max_new_tokens=512)
    choice = _extract_mcq_choice(raw, options)
    if choice is not None:
        return choice

    # Retry
    retry_prompt = f"""Question:
{question}

Options:
{options_text}

Show your work step by step, then state your final answer.
ANSWER: """
    raw = _generate_mcq(retry_prompt, max_new_tokens=384)
    return _extract_mcq_choice(raw, options)
```

In [82]:

```
rows_raw = []

for c in tqdm(cases, desc="Evaluating RAW model"):
    pred = answer_mcq_raw_model(c["question"], c["options"])
    rows_raw.append({
        "topic": c.get("topic"),
        "module": c.get("module"),
        "mcq_id": c.get("mcq_id"),
        "question_number": c.get("question_number"),
        "question": c["question"],
        "gold": c["correct_answer"],
        "pred": pred,
        "correct": pred == c["correct_answer"],
    })
```

```
Evaluating RAW model:   0%|          | 0/28 [00:00<?, ?it/s]
2026-03-11T18:42:49Z INF Lost connection with the edge connIndex=2
2026-03-11T18:42:49Z ERR failed to serve incoming request error="Error shutting down control stream: context canceled"
2026-03-11T18:42:49Z WRN Serve tunnel error error="connection with edge closed" connIndex=2 event=0 ip=198.41.200.113
2026-03-11T18:42:49Z INF Retrying connection in up to 1s connIndex=2 event=0 ip=198.41.200.113
2026-03-11T18:42:49Z INF Lost connection with the edge connIndex=1
2026-03-11T18:42:49Z WRN Serve tunnel error error="connection with edge closed" connIndex=1 event=0 ip=198.41.200.53
2026-03-11T18:42:49Z INF Retrying connection in up to 1s connIndex=1 event=0 ip=198.41.200.53
2026-03-11T18:42:49Z ERR failed to serve incoming request error="Error shutting down control stream: client disconnected"
2026-03-11T18:42:50Z WRN Connection terminated error="connection with edge closed" connIndex=1
2026-03-11T18:42:50Z WRN Connection terminated error="connection with edge closed" connIndex=2
2026-03-11T18:42:53Z INF Lost connection with the edge connIndex=2
2026-03-11T18:42:53Z WRN Serve tunnel error error="connection with edge closed" connIndex=2 event=0 ip=198.41.200.73
2026-03-11T18:42:53Z INF Retrying connection in up to 1s connIndex=2 event=0 ip=198.41.200.73
2026-03-11T18:42:53Z ERR failed to serve incoming request error="Error shutting down control stream: context canceled"
2026-03-11T18:42:53Z INF Lost connection with the edge connIndex=0
2026-03-11T18:42:53Z WRN Serve tunnel error error="connection with edge closed" connIndex=0 event=0 ip=198.41.200.233
2026-03-11T18:42:53Z INF Retrying connection in up to 1s connIndex=0 event=0 ip=198.41.200.233
2026-03-11T18:42:53Z ERR failed to serve incoming request error="Error shutting down control stream: context canceled"
2026-03-11T18:42:53Z WRN Connection terminated error="connection with edge closed" connIndex=2
2026-03-11T18:42:54Z WRN Connection terminated error="connection with edge closed" connIndex=0
2026-03-11T18:43:00Z INF Registered tunnel connection connIndex=2 connection=7655d9af-46ec-4867-9d69-f6b7aab0b21e event=0 ip=198.41.200.113 location=ord12 protocol=h
ttp2
2026-03-11T18:43:00Z INF Registered tunnel connection connIndex=0 connection=3ec47bf2-825f-47aa-b0a7-0ea82d59b5b1 event=0 ip=198.41.200.233 location=ord11 protocol=h
ttp2
2026-03-11T18:43:00Z INF Registered tunnel connection connIndex=1 connection=9a697f14-8697-4b74-a590-9477d48e9865 event=0 ip=198.41.200.53 location=ord12 protocol=ht
tp2
2026-03-11T18:43:01Z INF Registered tunnel connection connIndex=2 connection=2942ae1c-26aa-40fc-9225-91ea39e5d1b8 event=0 ip=198.41.200.73 location=ord11 protocol=ht
tp2
```

In [83]:

```
df = pd.DataFrame(rows_raw)

print("Overall accuracy:", df["correct"].mean())
print("Correct:", df["correct"].sum(), "/", len(df))

print()
print("By topic:")
print(
    df.groupby("topic", dropna=False)
      .agg(
          n=("correct", "size"),
          correct=("correct", "sum"),
          accuracy=("correct", "mean"),
      )
      .sort_values("topic")
)
```

```
Overall accuracy: 0.4642857142857143
Correct: 13 / 28

By topic:
                                   n  correct  accuracy
topic
gift_estate_and_charitable         7        3  0.428571
passive_activity_and_at_risk       7        2  0.285714
property_transactions              7        5  0.714286
retirement_and_financial_planning  7        3  0.428571
```

Reload Retrieval Stack

In [84]:

```
import gc, torch
from sentence_transformers import SentenceTransformer, CrossEncoder
from pymilvus import MilvusClient

# Reload embedder
if 'embedder' not in dir() or embedder is None:
    embedder = SentenceTransformer(EMBED_MODEL_NAME)
    print("Embedder reloaded.")
else:
    print("Embedder already loaded.")

# Reload reranker
if 'reranker' not in dir() or reranker is None:
    reranker = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2", max_length=512)
    print("Reranker reloaded.")
else:
    print("Reranker already loaded.")

# Reconnect Milvus
try:
    client.close()
except Exception:
    pass
client = MilvusClient(DB_PATH)
print("Milvus client reconnected.")

# Quick sanity check
cols = client.list_collections()
print(f"Collections: {cols}")
print("Retrieval stack is live — API and interactive cells will work.")
```

```
Embedder reloaded.
Reranker reloaded.
Milvus client reconnected.
Collections: ['irs_individual']
Retrieval stack is live — API and interactive cells will work.
```

In [84]: